

# Comprehensive glossary covering all the core mathematical, statistical, and machine learning concepts in detail:

## 1. Linear Algebra Fundamentals

\* **Matrix Multiplication ( $C = AB$ ):** An algebraic operation where two matrices are multiplied to produce a third matrix. For the product to exist, the inner dimensions must match (the number of columns in A must equal the number of rows in B). Each element in C is computed via the dot product of a row from A and a column from B.

\* **Non-Commutativity ( $AB \neq BA$ ):** The rule that the order of matrix multiplication matters. Reversing the order typically changes the resulting matrix or makes the operation mathematically invalid. Geometrically, this reflects that applying sequential linear transformations (like scaling and rotation) in a different order yields a different final state.

\* **Gaussian Elimination:** A systematic algorithmic process used to solve systems of linear equations ( $Ax = b$ ). It applies elementary row operations (row swapping, multiplication, and addition) to transform an augmented matrix into row-echelon form or reduced row-echelon form.

\* **Vector Space ( $V = (\mathcal{V}, +, \cdot)$ ):** A formal mathematical structure consisting of a set of vectors  $\mathcal{V}$  equipped with two operations: vector addition and scalar multiplication. These operations must satisfy specific axioms, such as associativity, commutativity, and the presence of additive identities and inverses.

\* **Linear Independence:** A property of a set of vectors indicating that no vector in the set can be expressed as a linear combination of the others. Mathematically, a set of vectors is linearly independent if the equation  $\sum \lambda_i x_i = 0$  holds true exclusively when all scalar coefficients  $\lambda_i$  are zero.

\* **Matrix Inverse ( $A^{-1}$ ):** A unique square matrix associated with a given invertible matrix A such that multiplying them yields the identity matrix ( $AA^{-1} = A^{-1}A = I$ ). It effectively "undoes" the linear transformation performed by A.

\* **Eigendecomposition ( $Av = \lambda v$ ):** A factorization of a square matrix into canonical forms involving its eigenvalues ( $\lambda$ ) and eigenvectors ( $v$ ). Geometrically, an eigenvector represents a direction that is unaffected (except for scaling by  $\lambda$ ) when the matrix transformation is applied.

\* **Singular Value Decomposition (SVD):** A powerful matrix factorization method ( $X = U\sigma V^T$ ) applicable to any rectangular matrix. Unlike eigendecomposition (which requires square matrices), SVD decomposes a matrix into orthogonal vectors and singular values, making it invaluable for dimensionality reduction, noise filtering, and data compression.

## 2. Optimization and Regression

\* **Normal Equation ( $\theta = (X^T X)^{-1} X^T y$ ):** An analytical, closed-form solution used in Ordinary Least Squares (OLS) linear regression. It calculates the optimal parameter vector ( $\theta$ ) that minimizes the mean squared error directly in a single mathematical step, bypassing iterative training.

\* **Mean Squared Error (MSE):** A standard cost function  $J(\theta) = 1/2m \sum (h_{\theta}(x^{(i)}) - y^{(i)})^2$  that measures the average squared difference between a model's predicted values and the actual target values. Squaring the residuals heavily penalizes larger errors.

\* **Gradient Descent ( $\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla J(\theta)$ ):** An iterative optimization algorithm used to find the minimum of a cost function. It updates parameters step-by-step in the exact opposite direction of the gradient vector, scaled by a learning rate ( $\alpha$ ).

\* **RMSprop:** An advanced gradient-based optimization algorithm designed for neural networks. It resolves the problem of diminishing or exploding gradients by dividing the learning rate by an

exponential moving average of recent squared gradients, adapting step sizes per parameter.

\* **Adam (Adaptive Moment Estimation)**: A widely used optimization algorithm that merges the best properties of both AdaGrad/RMSprop (handling sparse gradients via squared past gradients, or second moments) and Momentum (incorporating exponential moving averages of past gradients, or first moments), complete with bias correction terms.

### 3. Neural Networks and Deep Learning

\* **Categorical Cross-Entropy Loss** ( $L = -\sum y_{o,c} \log(p_{o,c})$ ): A loss function used primarily for multi-class classification problems. It measures the dissimilarity between the true probability distribution (represented as one-hot encoded vectors) and the predicted probability distribution output by the model.

\* **Scaled Dot-Product Attention** ( $\text{Attention}(Q,K,V) = \text{softmax}(QK^T/\sqrt{d_k})V$ ): The fundamental mechanism behind Transformer architectures. It computes compatibility scores between queries (Q) and keys (K), scales them by the square root of the key dimension ( $\sqrt{d_k}$ ) to stabilize gradients, applies a softmax function, and uses the weights to aggregate values (V).

\* **Low-Rank Adaptation (LoRA)**: A parameter-efficient fine-tuning technique for large language models. Instead of updating all billions of parameters, LoRA freezes the original weight matrix ( $W_0$ ) and trains two smaller, low-rank decomposition matrices (B and A) to approximate the necessary weight updates ( $h = W_0x + BAx$ ).

\* **Sigmoid Function** ( $g(z) = 1/(1+e^{-z})$ ): A classic S-shaped non-linear activation function that squashes any real-valued number into a smooth range between 0 and 1, historically used for mapping outputs to probabilities.

\* **Backpropagation Error Term** ( $\delta^{(L)} = \nabla_a L \odot \sigma'(z^{(L)})$ ): The fundamental mathematical expression used during the backward pass of a neural network to calculate the error signal at the output layer, combining the partial derivative of the loss with respect to the activations, combined via the Hadamard product ( $\odot$ ) with the derivative of the activation function.

### 4. Information Theory and Ensembles

\* **Entropy** ( $H(S) = -\sum p_c \log_2(p_c)$ ): A core concept from information theory that quantifies the level of impurity, disorder, or uncertainty within a dataset across various classes (c).

\* **Gini Impurity** ( $G(S) = 1 - \sum p_c^2$ ): A computationally efficient metric used in decision trees to measure how often a randomly chosen element from a set would be incorrectly labeled if it were randomly labeled according to the distribution of labels in the subset.

\* **Information Gain** ( $IG(S,A)$ ): The expected reduction in dataset entropy achieved by partitioning the data based on a specific attribute or feature (A). Decision trees use information gain to select the best split criteria at each node.

\* **AdaBoost Weighted Error** ( $\epsilon_i$ ): The weighted error rate of a weak classification model within an AdaBoost ensemble. It is calculated by summing the weights of the training instances that the weak learner misclassified.

\* **Random Forest Variance Reduction** ( $\text{Var} = \rho\sigma^2 + 1 - \rho/B\sigma^2$ ): A statistical principle showing how averaging multiple randomized decision trees (B) decreases overall model variance. As the number of trees grows, the variance drops, provided the correlation ( $\rho$ ) between individual trees remains low.

### 5. Probability and Reinforcement Learning

\* **Bayes' Theorem** ( $P(h|D) = P(D|h)P(h)/P(D)$ ): A foundational theorem of probability calculus that provides a principled way to update the probability of a hypothesis (h) as more evidence or data (D) becomes available.

- \* **Gaussian (Normal) Distribution:** A continuous probability density function characterized by a symmetric, bell-shaped curve. It is completely defined by two parameters: its mean ( $\mu$ ) governing its central location, and its variance ( $\sigma^2$ ) governing its spread.
- \* **Expected Return ( $G_t$ ):** The core optimization target in reinforcement learning, representing the total sum of discounted rewards an agent expects to accumulate starting from time step  $t$  onward ( $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$ ), where  $\gamma$  is the discount factor.
- \* **Q-Learning Update Rule:** A model-free, off-policy temporal difference reinforcement learning algorithm. It iteratively updates an agent's estimate of the action-value function ( $Q(s,a)$ ) based on the reward received and the maximum anticipated reward of the next state.